

Justification of the 4D Torus Geometry

and Derivation of L_0 as the Smallest Stable Winding

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<https://github.com/jpascher/T0-Time-Mass-Duality>

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- Series: "on a circle the harmonic series converges" $\rightarrow \sum 1/n$ diverges, $\sum 1/n^2 = \pi^2/6$ converges, both independent of line or circle; torus overtones $m_n = m_0 n^{1/3}$ and "the Koide formula follows" annotated ($m_\mu/m_e = 206.8$ requires $n \approx 8.8 \cdot 10^6$); redshift $z \approx \xi \ln(d/\ell_P)$ annotated ($z = 0.019$ at $d = R_H$).
- (2 Oct. 2026) Notation f : base winding number $f = 1/\xi$ — not a frequency; note in the text (Doc. 190, R147).

Abstract

The 4D torus is not a postulate of T0 theory. It follows necessarily from three independent conditions: (1) freedom from divergences in quantum field theory requires periodic topology, (2) the value $\xi = 4/30000$ encodes exactly four spacetime dimensions, (3) stable winding numbers on the torus generate the observed quantum numbers. The minimal length scale $L_0 = \xi \cdot \ell_P$ is geometrically the length of a single torus lattice cell — the smallest stable winding. The Lagrangian and the torus yield L_0 consistently via two independent paths.

.1 Why a Fourth Dimension?**The Divergence Problem of Quantum Field Theory**

Quantum field theory suffers from a fundamental problem: the vacuum energy diverges:

$$E_{\text{vac}} = \sum_n \frac{1}{2} \hbar \omega_n \rightarrow \infty \quad (1)$$

The observed cosmological constant is 10^{120} times smaller than the QFT prediction. The Standard Model resolves this through artificial renormalisation. T0/FPGFT resolves it geometrically.

Periodic Geometry as Solution

The harmonic series $\sum 1/n$ diverges; the series of inverse squares converges — Euler 1735 (*Corrected on 30 Sept. 2026: previously "On an infinite line the harmonic series diverges. On a circle it converges" — $\sum 1/n$ diverges and $\sum 1/n^2 = \pi^2/6$ converges, both regardless of whether the modes are pictured on a line or a circle; the topology does not turn the divergent series into a convergent one.*):

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \quad (2)$$

Topological Conclusion

The periodic geometry automatically suppresses high modes. If spacetime has a periodic topology, UV divergences vanish geometrically — without artificial renormalisation. The π^2 in Euler's result is not number theory, but the signature of the topology.

Four Dimensions from ξ

The parameter $\xi = 4/30000$ encodes the dimensionality of spacetime:

$$\xi_d \sim (10^{-1})^d \Rightarrow \text{for } d = 4 : \quad \xi_4 \sim 10^{-4} \quad (3)$$

Four Dimensions from ξ

The **4** in the numerator of $\xi = 4/30000$ stands directly for the four spacetime dimensions. The fourth dimension is not postulated — it is already contained in the value of ξ .

.2 Why a Torus?

Winding Numbers as Conserved Quantities

On a torus, winding numbers are topologically conserved. They generate stable discrete states:

$$\text{Spin} = \text{winding number } w = n_\phi / n_\theta \quad (4)$$

$$\text{Charge} = \text{flux } \Phi = n \cdot h/e \quad (5)$$

$$\text{Masses} = \text{torus overtone frequencies} \quad (6)$$

Torus vs. Sphere

A sphere S^3 is simply connected — no stable winding numbers. The torus is not simply connected — it generates exactly the observed quantum numbers. This is the topological reason for choosing the torus.

Torus Eigenvalues and Particle Masses

The Laplace operator on a d -dimensional torus with radii R_i has eigenvalues:

$$E_{\vec{n}} = \frac{\hbar^2}{2} \sum_i \frac{n_i^2}{R_i^2}, \quad \vec{n} \in \mathbb{Z}^d \quad (7)$$

Particle masses appear as torus overtone frequencies with step size $1/3 = 1/D_f$:

$$m_n = m_0 \cdot n^{1/3} \quad (8)$$

The Koide formula for lepton masses follows as a geometric consequence. (*Note of 30 Sept. 2026: With $m_n = m_0 n^{1/3}$ the first overtones $n = 2, 3$ give the ratios 1.26 and 1.44; $m_\mu/m_e = 206.8$ requires $n = 206.8^3 \approx 8.8 \cdot 10^6$. A derivation of the Koide formula is not given here.*)

Compactification

The fourth dimension is rolled up to radius r_4 . This radius must coincide with the sub-Planck scale from the Lagrangian — otherwise the theory would be inconsistent. The condition:

$$r_4 = L_0 = \xi \cdot \ell_P \quad (9)$$

The topology is therefore:

$$\mathbb{R}^3 \times S^1 \quad \text{with} \quad r_4 = \xi \cdot \ell_P \quad (10)$$

.3 Derivation of L_0 from Torus Geometry

The Winding Density

The torus has $f = 1/\xi$ sub-Planck cells per Planck length. With $\xi = 4/30000$:

$$f = \frac{1}{\xi} = \frac{30000}{4} = 7500 \quad \text{cells per } \ell_P \quad (11)$$

This number is not arbitrary — it follows directly from the geometric winding density $30000/4$ of the torus. The prime factorisation $7500 = 2^2 \times 3 \times 5^4$ encodes the symmetry structure of spacetime.

(Note of 2 Oct. 2026 on notation: here f is the base winding number $f = 1/\xi = 7500$, not a frequency. In natural units f is only another reading of the duality $T \cdot m = 1$: $f = t_P/t_0 = \Lambda_{T0}/E_P$ with $t_0 = \xi t_P$ and $\Lambda_{T0} = E_P/\xi$, and $\xi E_0^2 = 1$ gives $f = E_0^2$. As a dimensionless ratio, f cannot be set to 1 with justification (Doc. A135). Cf. Doc. 190, R147.)

Smallest Stable Winding

The smallest stable winding on the torus has the length of a single lattice cell:

$$L_0 = \frac{\ell_P}{f} = \ell_P \cdot \xi \quad (12)$$

Numerically:

$$L_0 = 1.616 \times 10^{-35} \text{ m} \times \frac{4}{30000} = 1.616 \times 10^{-35} \text{ m} \times 1.333 \times 10^{-4} \quad (13)$$

Minimal Length Scale from Torus Geometry

$$L_0 = \xi \cdot \ell_P = 2.155 \times 10^{-39} \text{ m} \quad (14)$$

L_0 is the length of a single torus lattice cell — the smallest stable winding.

Geometric Interpretation

L_0 is not simply ξ times the Planck length — L_0 is the length of a single torus lattice cell. Below L_0 :

- The concept of "distance" loses its physical meaning.
- All vacuum fluctuations are fully effective.
- The fractal correction K_{frak} becomes structurally necessary.

Consistency of the Two Derivation Paths

L_0 is not derived twice — the Lagrangian and the torus yield the same number via two independent paths:

Two Consistent Paths to L_0

$$\begin{aligned}
 \text{Lagrangian: } g_T = \xi \cdot m &\Rightarrow \xi = 4/30000 \\
 \Downarrow \\
 \text{Torus: } f = 1/\xi &= 7500 \text{ cells per } \ell_P \\
 \Downarrow \\
 \text{Smallest winding: } L_0 = \ell_P / f &= \xi \cdot \ell_P \\
 \Downarrow \\
 \boxed{L_0 = 2.155 \times 10^{-39} \text{ m}} &\quad (\text{consistent from both paths}) \quad (15)
 \end{aligned}$$

The Lagrangian provides ξ . The torus explains what ξ means geometrically: the reciprocal winding density.

.4 Comparison with Other Approaches

String Theory

String theory requires 10 or 11 dimensions. The T0 Lagrangian with a single parameter ξ suffices with 4. Every additional dimension would be a postulate — in FFGFT there are no free parameters.

Loop Quantum Gravity

LQG discretises space into spin networks. T0 discretises spacetime through the periodic torus geometry. The difference: in T0 the discreteness is a topological consequence, not an additional postulate.

Asymptotic Safety

AS seeks a UV fixed point as the result of renormalisation group flow. In T0 there is no renormalisation group flow — ξ is a geometric constant, not a running parameter. The fixed point is not sought but established geometrically from the outset.

.5 Physical Consequences

Vacuum Catastrophe Resolved

On the fractal torus, the vacuum energy converges through periodic suppression of high frequencies. No artificial renormalisation required.

Three Generations as Overtone Structure

Electron, muon and tau are not three independent particles — they are the fundamental tone and first overtones of a single torus resonator. The hierarchy follows from the torus eigenvalues:

$$m_e : m_\mu : m_\tau \approx 1 : 206.8 : 3477 \quad (16)$$

No Big Bang and No Frequency Dependence

The torus has no boundaries. The observed redshift does not arise through expansion but through geometric energy loss — fractal summation extends the effective path length of photons through the T0 vacuum field:

$$z \approx \xi \cdot \ln\left(\frac{d}{\ell_P}\right) \quad (17)$$

(Note of 30 Sept. 2026: For $d = R_H = 1.4 \cdot 10^{26} \text{ m}$, $\ln(d/\ell_P) = 140.3$ and hence $z = \xi \cdot 140.3 = 0.019$; $z \approx 10$ would require $\ln(d/\ell_P) = 75\,000$. In this form the formula is incompatible with the observed redshifts $z > 1$.)

Frequency Independence of the Redshift

Since the path extension is a property of the spacetime geometry itself, red and blue photons starting from the same location take exactly the same extended path. Their wavelengths are therefore stretched by the same percentage — there is no frequency dependence. This distinguishes the T0 mechanism fundamentally from simple “tired light” models, which fail precisely on the observed frequency independence. Derived in detail in Document 026 (Geometric Cosmology).

.6 Summary

Three Independent Conditions Point to the Same Torus

- (1) Freedom from divergences $\Rightarrow$ periodic topology
- (2) $\xi = 4/30000 \Rightarrow$ exactly 4 dimensions
- (3) Winding numbers $\Rightarrow$ stable quantum numbers

All three point to the same conclusion:

$$\mathbb{R}^3 \times S^1 \quad \text{with} \quad r_4 = L_0 = \xi \cdot \ell_P = 2.155 \times 10^{-39} \text{ m} \quad (18)$$

Note: Connection to the SI System

The absolute numerical values in this document — in particular $L_0 = 2.155 \times 10^{-39} \text{ m}$ — presuppose a connection to the SI unit system. This is not circular but epistemologically necessary: from a dimensionless parameter ξ alone, no absolute metre values can be derived. The connection is made via m_e , ℓ_P or α as equivalent entry points. Treated in detail in Document 180 (Section: Connection to the SI System) and in Documents 056 and 101.

Documents 180, 181 and 182 form a unit and should not be read separately.